

Linear algebra: Exercise Chapter 8

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1. Determine which one of following matrix is diagonalizable and diagonalize them. Compute algebraic and geometric multiplicity for each eigenvalues.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 3 & -19 & 11 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

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2. Consider the matrix A in question 1. Compute A^5 .

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3. Suppose A is a diagonalizable matrix with $A = X\Lambda X^{-1}$. Diagonalize $A + 2I$.

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4. Let A, B and C be $n \times n$ matrices. Prove the following statements:

- If A is similar to B and B is similar to C , then A is similar to C .
- If A is similar to B , then A^2 is similar to B^2 .
- If A and B are similar matrices, then their determinants are the same.

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5. True or false: If the columns of matrix X are the eigenvectors of a matrix A and they are linearly independent, then

- (a) A is invertible;
- (b) A is diagonalizable;
- (c) X is invertible;
- (d) X is diagonalizable.

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6. True or false: If the only eigenvectors of A are multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, then

- (a) A is not invertible;
- (b) A has a repeated eigenvalue;
- (c) A is nondiagonalizable.